RISHABADEV P

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SUMMARY

Engineering student specializing in **AI/ML, robotics, and embedded systems**, with experience in computer vision, sensor integration, and real-time embedded applications. Skilled at rapidly prototyping hardware–software solutions and developing practical systems that bridge intelligent algorithms with physical devices.

EDUCATION

B.Tech in Electronics and Communication

Engineering -(2023)

Cochin University of Science and Technology (CUSAT)

SKILLS

Core: Hardware & Embedded Systems, Circuit Design, Sensor Integration, Prototyping
Programming: Python(Django), C/C++,HTML,CSS, MATLAB

AI/ML & Vision: OpenCV, Scikit-learn,, Model CNN,Tensor-Flow,Training
Tools: ESP32, Arduino, Vivado, Proteus, KiCAD, VS Code

.PROJECTS

• **Pollution Mapping System**
Developed a **low-cost mobile air quality monitoring system using ESP32** to measure particulate matter ($PM_{2.5}$), temperature, and humidity.
Integrated a **GNSS module for geolocation** to associate pollution measurements with geographic coordinates.
Implemented **Wi-Fi data transmission and microSD logging** for real-time monitoring and redundant data storage.
Visualized collected environmental data on a **web-based map to analyze spatial air pollution patterns**.
GitHub: https://github.com/Rishabadev/pollution-mapper

• **Smart Handwritten Digit Recognizer**
Developed a **machine learning-based handwritten digit recognition system** using image processing and classification techniques.
Implemented preprocessing methods including **image normalization, contour extraction, and digit segmentation**.
Designed a pipeline capable of **predicting digits drawn by the user in real time** with visual feedback.
GitHub: https://github.com/Rishabadev/Smart-Digit-Recognizer

• **Pothole Mapping System**
Developed a **low-cost road condition monitoring system using smartphone sensors and GPS**.
Detected potholes by identifying **accelerometer spikes corresponding to road irregularities**.
Uploaded detected locations to a backend server and **visualized pothole coordinates on an interactive map using Leaflet.js**.

• **Binary Floating Point Adder (Digital Design Project)**
Designed and simulated a **custom floating-point adder with a 4-bit mantissa and 3-bit exponent architecture**.
Implemented exponent alignment, mantissa addition, normalization, and overflow handling using digital logic modules.
Validated arithmetic functionality using **Proteus simulation and HDL implementation in Vivado**.

• **Computer Vision Image Processing Toolkit**
Developed a **GUI-based image processing tool using Python, OpenCV, and Tkinter** for performing real-time transformations.
Implemented features including **affine transformations, contour detection, rotation, scaling, and image warping**.

• **Django eCommerce Web Application**
Built a **full-stack eCommerce platform using Django** with authentication, product management, and cart functionality.